07. Model Training & Hyperparameter Tuning

Follow the model-building code and identify what was actually tuned.

Project: Movie Recommender System

Source archive supplied in this conversation: Movie-recommender-system-main.zip | Notebook: notebook.ipynb | App: app.py

Training/build sequence

- Merge movies and credits by title.
- Select movie_id, title, overview, genres, keywords, cast and crew; drop rows with nulls.
- Parse genres/keywords/cast into names and crew into directors.
- Retain the first three cast members and remove spaces from multi-word names.
- Split overview into tokens and concatenate all fields into tags.
- Fit CountVectorizer(max_features=5000, stop_words='english') on all movie tags.
- Compute cosine_similarity(vector), yielding a dense pairwise item-item matrix.
- Save the processed movie table and similarity matrix with pickle.

Important parameter choices

Parameter	Repository value	Effect
max_features	5,000	Caps vocabulary size to the top terms selected by CountVectorizer
stop_words	english	Removes common English stop words
cast length	3	Uses first three cast entries
ranking depth	5	Returns five movies after excluding the selected movie
similarity	cosine	Ranks vectors by angular similarity

Is there hyperparameter tuning?

No systematic tuning is present. The vocabulary cap, stop-word setting and cast limit are fixed. There is no experiment over max_features, min_df, n-grams, field weights or TF-IDF, and no ranking metric to select the best configuration.

Complexity and scaling

With N movies and V vocabulary terms, the feature matrix is approximately N×V. A dense similarity matrix is N×N, requiring $O(N^2)$ storage and pairwise computation. For 4,806 movies this is manageable, but memory grows quadratically. At larger catalogue sizes, use sparse nearest-neighbour search, approximate nearest neighbours, vector databases or on-demand top-K retrieval rather than storing every pairwise score.

Persistence caveats

The notebook saves new (movie_id, title, tags) and similarity using pickle. The app expects them under model/movie_list.pkl and model/similarity.pkl, but the notebook saves them in the current working directory; the supplied ZIP does not include a model/ folder or generated pickle files. The file paths must be aligned when reproducing the app. Only load pickle files from

trusted sources.